

Rooted Resilience: Natural Farming for Climate Resilience and Livelihood Security of PVTG Households in Durgakondal Block, Uttar Bastar Kanker, Chhattisgarh

A three-year Climate Smart Livelihoods proposal submitted by EcoKheti Foundation to SBI Foundation (SBIF). The programme reaches up to 1,000 Particularly Vulnerable Tribal Group (PVTG) households through a women-led community resource person model converged with the National Mission on Natural Farming (NMNF), the Forest Rights Act (FRA) and PM-JANMAN. Request: INR 99,00,000 (costed at INR 98,89,000), over 12 quarters starting in the current financial year (inferred as FY 2025-26, to be confirmed with SBIF).

Location: Durgakondal (Durgkondal) Block, Uttar Bastar Kanker District, Chhattisgarh

Duration: 3 years

Budget: INR 99,00,000

Submitted to: SBI Foundation (SBIF)

Submitted by: EcoKheti Foundation

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Problem Statement

This chapter sets out why PVTG households in Durgakondal block face insecure livelihoods and rising climate exposure, and why that situation will not correct itself. It looks at the problem from each angle the available evidence allows. It is also candid about what is not yet known, because an honest account of the evidence gaps shapes the design as much as the known facts do. The picture that emerges has several layers. Remoteness raises the cost of every service. A deeply tribal population sits at the margins of state systems. Land tenure is still being settled. Low literacy, especially among women, limits access to both knowledge and entitlements. No local evidence yet shows which farming pathway would work.

Geographic and administrative context: a remote, sparse block

Durgakondal tehsil had 64,293 people in 12,888 households spread over about 900 square kilometres, a density of only 71 persons per square kilometre ^[S1]. That density has practical consequences. Habitations are small and far apart, and most of the block is forested. The unit cost of reaching each household is therefore high for an extension worker, a veterinary service, an input dealer or a scheme camp. Services cluster where roads and people are densest, so the most remote hamlets are reached last and least often.

Children aged 0 to 6 made up 15.11% of the population ^[S1]. This is a young, dependent population, and it means that any shock to farm income passes quickly into the food and care of small children. These Census 2011 figures come from a secondary aggregator and will be checked against the District Census Handbook. The geography itself argues for a delivery model based inside the community, rather than one that relies on visiting staff.

Social and community structure: a deeply tribal block with PVTG pockets

Scheduled Tribes number 49,250 people in the block, about 76.6% of the population, against 55% for Kanker district as a whole ^{[S2][S3]}. Durgakondal is therefore considerably more tribal than its own district, and programmes designed for the district average will fit it poorly.

Within this tribal population, PVTGs are the most vulnerable. Chhattisgarh has 5 centrally notified PVTGs, with a combined population of 1,86,918 across 18 districts, and Kanker was a focus of the Election Commission's PVTG voter outreach ^[S5]. The fact that PVTGs needed dedicated outreach even to be brought into the electoral process shows how far they sit from routine state contact. The Kamar PVTG are recorded in Kanker, along with Gariaband, Dhamtari and Mahasamund, with a 2011 population of 26,630. The state has also given PVTG status to Pando and Bhunjia ^[S6].

It is not yet known which PVTG or PVTGs live in Durgakondal, or how many households. No block-level PVTG count was found. This is itself a symptom of the problem, because a group that is not counted at habitation level cannot be targeted reliably. The project therefore begins with verification through PM-JANMAN habitation data and a field listing.

Land tenure: rights in flux on forest land

For forest-dwelling households, the question of whose land it is comes before the question of how to farm it. Chhattisgarh's FRA process is large. As of September 2021 it had received 9,17,690 claims and issued 4,90,587 titles, covering about 8.98 lakh acres under individual rights and 47.79 lakh acres under community rights ^{[S8][S9]}. District and block figures are not available and will be collected.

The pattern points to a second-generation problem. A title secures the right to occupy land, but it does not by itself make that land productive or sustainably used. Households with newly titled forest land often lack the means, inputs or confidence to invest in it.

For PVTGs, tenure is still evolving. The first PVTG habitat rights in the state went to the Kamar in 22 settlements of Magarlod, Dhamtari, in 2023, and a single Kamar habitat title covering Kanker is planned ^{[S6][S7]}. This is an opening. If households in Durgakondal fall within that process, farming improvements can be placed on land that is becoming legally secure. Without facilitation, however, eligible households in remote habitations risk being missed.

Economic and livelihoods dimension: a system not yet measured

The block's cropping pattern, irrigation, landholding size, yields, chemical input use and cash input costs are not yet documented. Nor is the degree of dependence on forest produce. The baseline survey will establish all of these. Field data will show how far cash spending on purchased inputs eats into margins here.

The logic of the problem can still be reasoned through. Where households farm small plots on uncertain land with little access to credit, every rupee spent on purchased inputs raises both risk and exposure to debt. A bad season then leaves a household with a lost crop and an unpaid cost. External evidence shows how powerful input costs can be. In Andhra Pradesh's natural farming programme, including in tribal systems, the gains in net income came mainly from lower input costs ^[S10]. A pathway that turns cash inputs into inputs made on the farm addresses this exposure directly.

Agro-ecological and climate dimension

Local rainfall trends and dry spells have not yet been compiled. IMD data will be analysed as part of the baseline. What is clear is the direction of risk. A forested, rainfed landscape depends on the timing and amount of rain, and farming systems built around one or two crops have little to fall back on when a season fails.

The Government of India's position is that natural farming cuts purchased inputs, restores soil health and builds resilience to drought, flood and waterlogging ^[S13]. In Andhra Pradesh, crop diversity rose from 2.1 to 4 crops per farm ^[S11]. Diversity is a form of insurance, because several crops are unlikely to fail together. For PVTG households, whose traditional systems likely included mixed cropping and indigenous varieties, this is a return to a resilient logic rather than a foreign imposition. The baseline will confirm how much of that system remains.

Education and human capital

Block literacy is 53.15%, and female literacy is 44.12%, against 70.29% for the district. NHSRC gives ST literacy in the district as 55% ^{[S1][S4][S3]}. These numbers bear directly on the design. Written extension leaflets, scheme application forms and classroom trainings will not reach most adults, and will reach women least of all. Knowledge that exists only on paper does not travel in Durgakondal. It has to be shown in the field, in the local language, by someone the community trusts.

The gender dimension

The gap between female literacy in the block (44.12%) and district literacy (70.29%) ^{[S1][S4][S3]} reflects a wider exclusion. Women in such settings typically do much of the farm and household labour, yet have the least access to training, documents and decision-making forums. Excluding women therefore weakens adoption itself, not only fairness. The Andhra Pradesh study found that women played a key role in the stronger social capital that natural farming built ^[S12]. A model that places women at the centre of knowledge transfer uses a strength that is currently wasted.

Health and nutrition

Block-level nutrition status will be drawn from NFHS-5 and the baseline, so no local claim is made here. The links still matter. With 15.11% of the population under the age of seven ^[S1], household food security bears directly on child growth. Farmers in Andhra Pradesh reported 33% fewer sick days and 26% lower health costs ^[S12]. Fewer sick days mean more working days, and lower health spending means less cash drawn out of the farm. A more diverse farm also supplies a more diverse diet, a link EcoKheti has seen in Khunti and Gumla, where dietary diversity improved in 60% of households.

Migration and its social cost

No data on seasonal migration from Durgakondal is available, and none is asserted here. The question still matters. When cash costs rise and seasons fail, households commonly look for wage work away from home, and this takes labour and care away from children and the farm. The baseline will measure outward labour movement so that the programme can track whether a more resilient farm reduces pressure to leave.

Access to government schemes and entitlements

A strong architecture of support already exists. NMNF offers farmer incentives and bio-input centre support. FRA offers tenure. PM-JANMAN targets PVTG habitations with a set of interventions. The 10,000 FPO scheme supports aggregation. The barrier is access. Remoteness, low literacy, uncertain PVTG enumeration and weak contact with the block administration all stand between these households and what they are entitled to. Entitlements that are not claimed produce no change, and each entitlement a household secures strengthens its capacity to secure the next.

The evidence gap and why the problem will not resolve on its own

No Chhattisgarh or Bastar field evidence on natural farming was found. The Andhra Pradesh evidence is encouraging. It surveyed 562 households across 3 agro-ecological zones between 2020 and 2022 and found 11% higher yields and 49% higher net incomes ^[S10]. It also found that labour use rose 20% ^[S11], which is a real design risk for PVTG households with little spare labour. Transferring that model without adapting it could leave households working harder for uncertain gain.

Each of these factors reinforces the others. Remoteness keeps services away. Low literacy keeps knowledge and entitlements out of reach. Insecure tenure discourages investment in land. Undiverse, cash-dependent farming raises exposure to climate shocks. None of these will ease through market forces or the routine working of schemes, because the households most affected are the ones least connected to both. The next chapter sets out the specific change this project seeks in response.

Project Objective

The problem analysis identified the core issues: farming that is exposed to cash costs and climate stress, land that is titled or being titled but underused, and entitlements that exist but go unclaimed. The objective answers each of these directly.

By the end of 3 years, up to 1,000 PVTG households in Durgakondal block practise natural farming on their own and FRA land. This lowers their cash input costs, diversifies crops and strengthens climate resilience and farm income. It is delivered through a women-led community resource person model converged with NMNE, FRA and PM-JANMAN.

Who. The cohort is up to 1,000 PVTG households. Because no block-level PVTG count exists, the number will be confirmed in Q1 through PM-JANMAN habitation data and a field listing. If fewer PVTG households are found, PVTG households will be enrolled first and the cohort topped up with other ST households.

Where. PVTG habitations of Durgakondal block, Uttar Bastar Kanker, Chhattisgarh. Villages will be identified in Q1.

When. 12 quarters over 3 years, starting in the current financial year (inferred as FY 2025-26, to be confirmed with SBIF).

What change. At least 600 households (60% of those enrolled) will practise at least 3 core natural farming practices on at least part of their land. Adopters are targeted to reach at least 20% lower cash input cost per acre, at least 25% higher net farm income and at least one additional crop per farm. These magnitudes will be validated after the baseline.

The objective also covers how the change is secured. Enrolment in NMNE, facilitation of FRA claims and linkage of Common Interest Groups (CIGs) to an FPO are part of the objective itself, because they are what allow the change to last. The strategy chapter explains why this combination is the right route.

Operational Strategy

This chapter explains the logic of the approach and why it fits Durgakondal. The strategy follows from the problem. Each design choice responds to a specific constraint identified in the analysis.

The theory of change

The theory of change connects four elements: secure land through FRA, low-cost natural farming, collective institutions, and entitlements. Together these lead to lower input costs and better climate resilience. The sequence matters. Natural farming reduces cash exposure and builds soil and crop diversity. Placing it on legally secure land gives households a reason to invest labour in that land over years. Collective institutions reduce the labour and cost of making inputs, and allow surplus to be sold on better terms. Entitlements bring in public resources the project could not fund alone, and they create roles and institutions that remain after the project closes. If any one element were removed, the others would weaken.

A women-led community resource person model aligned to NMNF

Last-mile delivery rests on 20 community resource persons (CRPs), preferably PVTG women, each supporting about 50 households. The model mirrors NMNF design, which provides for 2 Krishi Sakhis or CRPs per cluster and prefers clusters with existing SRLM groups and FPOs ^[S14] ^[S15]. This alignment is deliberate. CRPs will be registered as NMNF Krishi Sakhis so that the role continues after the project.

There are four reasons for this choice in Durgakondal. First, a density of 71 persons per square kilometre makes outside staff too costly and too infrequent, whereas a resident CRP is present every week. Second, a CRP speaks the local language and is trusted in ways an outsider is not. Third, preferring PVTG women builds women's agency. This draws on the Andhra Pradesh evidence of women's key role ^[S12] and on EcoKheti's SHG-led work with 1,200 women in Khunti and Gumla, where incomes rose by at least 45%. Fourth, it places knowledge within the community rather than with the implementing agency.

Designing for low literacy

With female literacy in the block at 44.12% ^[S1]^[S4]^[S3], all learning is practical. Farmer field schools take place in fields. Demonstration plots show results that can be seen and compared. Training is delivered in the local language. KVK and NMNF model-farm design informs the demonstration plots ^[S15]. Farmers learn by preparing Jeevamrit and Beejamrit with their own hands and by watching a neighbour's plot through a full season, not by reading.

Designing against the labour risk

The Andhra Pradesh finding that labour use rose 20% ^[S11] is taken seriously. For PVTG households with little spare labour, a practice that saves cash but adds drudgery may be dropped. The strategy responds in two ways. Practices are sequenced so that households begin

with the least labour-intensive ones. Inputs are also prepared jointly at CIG and Bio-input Resource Centre (BRC) level, so no household has to make every batch alone. This shift from individual to collective preparation reduces drudgery and also strengthens the group.

Convergence as the engine of scale and continuity

EcoKheti's values include convergence with public systems. Here convergence is a financing and sustainability strategy, not an add-on.

- **NMNF.** The farmer incentive of ₹4,000 per acre per year for 2 years goes directly to farmers. It is not a project cost, and the project facilitates enrolment ^[S14]. BRC set-up support of ₹1 lakh per BRC is sought from NMNF, and the project funds only shed and storage top-ups ^[S16].
- **FRA.** Claim facilitation, together with alignment to the planned Kamar habitat title covering Kanker, places natural farming on legally secure land ^{[S9][S7]}.
- **PM-JANMAN.** The project uses habitation-level PVTG data for targeting and links households to its 11 interventions, including housing, roads, water and livelihoods ^{[S17][S18]}.
- **FPO.** CIGs aggregate produce and link to an FPO under the 10,000 FPO scheme, which offers up to ₹18 lakh management support per FPO and ₹2,000 matching equity per member. These funds are paid to the FPO, not to the project ^[S19].

This approach adds public resources to SBIF's investment rather than duplicating them, and it leaves households tied into public systems that continue after the project.

Why EcoKheti

EcoKheti has worked in Jharkhand and Chhattisgarh since 2016. It brings 8 years in community-based livelihoods, 6 in maternal and child nutrition and 5 in climate-resilient livelihoods. In Chhattisgarh, 70% of 800 farmers it supported adopted climate-adaptive practices. In its FPO strengthening work with 3 FPOs and 900 members, aggregated sales roughly doubled. The strategy combines these three areas of proven capability: farm-level adoption, women-led institutions and collective marketing.

Managing strategic risk

NMNF is approved only until 2025-26. If it is not extended, CRP continuity will rely on Bihan (the Chhattisgarh SRLM) and on FPO revenue. The design keeps both pathways open from the start. A second risk is the lack of local evidence. The project answers this by generating Chhattisgarh-specific evidence through baseline, mid-term and endline measurement, and by validating targets once baseline data is available. The next chapter traces how these choices produce results at each level.

Activities

This chapter explains how each cluster of activities produces its result. The activities are sequenced over 12 quarters so that each one prepares the ground for the next, and together they build the theory of change.

A1: Targeting and baseline (Q1–Q2)

The first task is to establish who and where. PVTG habitations are verified against PM-JANMAN data, and up to 1,000 households are listed. The baseline then records agriculture, costs, income, crop diversity, FRA status and entitlements. Secondary collection adds IMD rainfall data, NFHS-5 nutrition data and NITI MPI poverty data.

This activity has three purposes. It turns an unverified target into a confirmed cohort. If fewer PVTG households are found, it triggers the top-up rule so that PVTG households come first. It also sets the baseline against which every outcome target is validated. The listing and baseline together with the endline are costed at ₹175 per household interview across 2,000 household-rounds. Enumerator pay must be at least the state minimum wage.

A2: Building the CRP cadre (Q1–Q11)

The CRP cadre is the core of the delivery model. Twenty CRPs, preferably PVTG women, are selected in Q1 and Q2. They complete a 6-day residential ToT, followed by four 2-day refreshers in Q5 and Q9, timed ahead of the seasons in which new practices are introduced. Residential training is costed at the MANAGE STRY/ATMA norm of ₹400 per trainee per day ^[S21] across 280 participant-days.

Each CRP receives a monthly honorarium for 30 months, from Q2 to Q11, at the BPKP norm of ₹5,000 per CRP per month ^[S20]. The honorarium is what allows a woman to commit regular time to about 50 households without giving up her own livelihood. Certification as an NMNF Krishi Sakhi gives the role a public identity. This is how Output 1.1 leads to Outcome 1: a resident, paid and certified extension worker in every cluster.

A3: Farmer field schools (Q3–Q10)

Each farmer attends 6 CRP-led sessions over two seasons, Kharif and Rabi. The sessions cover Jeevamrit, Beejamrit, mulching, multi-cropping, indigenous seed and pest management. They are held in the field and in the local language, which suits a block where female literacy is 44.12% ^{[S1][S4][S3]}. Spreading the sessions across two seasons lets farmers apply one practice, see its effect over a full cycle and return with questions before learning the next. Practices are introduced in order of labour demand, lightest first, in response to the labour risk identified in Andhra Pradesh ^[S11]. Sessions are costed at the BPKP rate of ₹100 per farmer per CRP-led village training ^[S20] across 6,000 farmer-sessions. Attendance registers and geo-tagged photos verify Output 1.2.

A4: Demonstration and shared infrastructure (Q2–Q11)

Seeing a practice work persuades more farmers than hearing about it. Each CRP maintains one model plot, 20 in total, designed on KVK and NMNF model-farm lines ^[S15], with signage and field days. Because the plot sits in the farmer's own village, the comparison is with local soil and local rain, not with a distant research farm.

Five BRC sheds are set up in Q3 to Q6 to supply bio-inputs to the community. NMNF BRC support of ₹1 lakh each is being sought ^[S16], and the project funds only the shed and storage top-up. Five community seed banks of indigenous varieties provide seed suited to local conditions and reduce the need to buy seed. The BRCs and seed banks supply the materials that turn training into continued practice, and they reduce the labour burden on individual households.

A5: Collectives and market linkage (Q2–Q12)

In Q2 to Q4, 100 CIGs of about 10 households each are formed and oriented, each with registers. Each CIG receives a shared starter kit of drums and tools. The CIG turns input preparation from a household task into a group task, which addresses the labour risk and builds the social capital that the Andhra Pradesh study linked to women's role ^[S12].

Exposure visits in Q4 and Q8 take 40 CRPs and lead farmers to working natural farming sites, so that they return with first-hand conviction to share with others. From Q7, once the first surpluses appear, CIGs aggregate produce and link to an FPO under the 10,000 FPO scheme, which supports the FPO directly ^[S19]. This draws on EcoKheti's experience with 3 FPOs, where aggregated sales roughly doubled. This activity cluster produces Outputs 2.1 and 2.2 and the collective marketing component of Outcome 2.

A6: Rights, entitlements and convergence (Q1–Q12)

Eight FRA claim-facilitation camps are held with the Tribal Department between Q2 and Q8. They cover documentation and GPS mapping, and they support habitat-rights processes, including alignment with the planned Kamar habitat title covering Kanker ^{[S9][S7]}. Households are enrolled in NMNF so that the ₹4,000 per acre per year incentive reaches them directly ^[S14], and are linked to PM-JANMAN interventions ^{[S17][S18]}. A block convergence meeting is held every quarter, 12 in all, to resolve bottlenecks with line departments. Holding them every quarter keeps the project's households in regular view of the block administration. This cluster produces Outputs 3.1 and 3.2 and, through them, Outcome 3.

A7: Monitoring, evaluation and learning (Q1–Q12)

Each CRP maintains a household tracker, which allows adoption to be tracked continuously rather than only at endline. The mid-term review in Q6 tests whether the labour-sequencing design and the CIG model are working, and allows course correction for the final seasons. The endline in Q11 and Q12 measures outcomes against baseline. Learning is documented, and Q12 is a planned handover. Because no Chhattisgarh or Bastar field evidence on natural

farming exists, this cluster produces more than accountability. It produces a public evidence base that others working in similar blocks can use.

Management and field presence

A Kanker-based Project Manager, a Natural Farming Specialist, two Field Coordinators (one of whom holds FRA and convergence duties), a part-time MEL and Data Officer and a part-time Accounts and Admin Assistant work from a field office in Durgakondal. Staff field travel is budgeted to support regular supervision across scattered habitations. The next chapter explains how the results of these activities are made to last.

Implementation Timeline

Activity	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
A1 PVTG verification, household listing and baseline												
A2 CRP selection, ToT and refreshers												
A2 CRP field support (honorary period)												
A5 CIG formation and starter kits												
A3 Farmer field schools (Kharif and Rabi)												
A4 Model demonstration plots												
A4 BRC sheds and community seed banks												
A6 FRA claim-facilitation camps												
A6 NMNF enrolment, PM-JANMAN linkage and convergence meetings												
A5 Exposure visits												
A5 CIG aggregation and FPO linkage												
A7 Mid-term review												
A7 Endline, documentation and handover												

Shaded cells indicate active quarters. Q1–Q2: mobilisation and group formation. Q3–Q6: capacity-building and asset support. Q7–Q8: consolidation, market linkage and transition.

Results Framework

This chapter follows the results chain from the ground up. It shows how outputs lead to outcomes and how outcomes add up to impact, so that each level visibly depends on the one below it.

Impact

Improved climate resilience and livelihood security of PVTG households in Durgakondal block, Kanker.

Resilience here means a farm that holds its output in a bad season. Security means a farm income that does not depend on buying expensive inputs on credit. The impact will be measured by the share of project households reporting stable or improved farm output in a climate-stress season, and by the change in household farm income. Both will be set against a baseline that the project will establish and validated using IMD seasonal data. The project claims an improvement over baseline among adopters, and will set the size of that improvement only after the baseline. It does this because no local evidence yet exists to support an earlier number.

Outcome 1: Adoption of natural farming

PVTG households adopt natural farming practices, including on-farm bio-inputs, crop diversification and indigenous seed, on their cultivated and FRA land.

Adoption is the starting point of the chain. None of the other gains can occur until practices change on the ground. The target is at least 600 households, 60% of those enrolled, practising at least 3 core practices on at least part of their land. This will be verified through the CRP household tracker, the endline survey and a field verification sample. The phrase "at least part of their land" is deliberate. Cautious farmers usually test on a portion of their land before converting all of it, and partial adoption that lasts is a better result than full adoption that is later abandoned.

This outcome follows from three outputs:

- **Output 1.1:** 20 CRPs trained and active. All 20 complete the 6-day ToT and refreshers, and at least 15 are certified as Krishi Sakhis.
- **Output 1.2:** At least 900 farmers attend at least 5 of the 6 field school sessions across Kharif and Rabi.
- **Output 1.3:** 20 demonstration plots, 5 community seed banks of indigenous varieties and 5 BRCs are functional.

The logic is straightforward. Trained CRPs deliver field schools. Field schools build skills. Demonstration plots give farmers visible proof. Seed banks and BRCs supply the materials needed to put the practices into use.

Outcome 2: Lower costs, higher income, greater diversity, collective marketing

Adopting households reduce cash cultivation costs and increase net farm income, crop diversity and collective marketing.

This outcome is the economic result of Outcome 1. Once bio-inputs are prepared on the farm or at the BRC, cash spending on purchased inputs falls. Multi-cropping raises the number of crops per farm. Collective sale of surplus improves the price realised. The targets are at least 20% lower cash input cost per acre, at least 25% higher net farm income and at least one additional crop per farm, all to be validated after the baseline. They will be measured through baseline, mid-term and endline surveys, supported by cost diaries kept by a sample of farmers. The Andhra Pradesh evidence, where net income gains came mainly from lower input costs ^[S10], supports this direction of change. The Durgakondal data will test how large the change is locally.

The outputs that drive this outcome are:

- **Output 2.1:** 100 CIGs, each with registers and shared kits in use. Shared kits lower each household's cost and labour, and the group becomes the unit for joint input preparation.
- **Output 2.2:** 40 participants complete exposure visits, and 100 CIGs are linked to an FPO for aggregation. Exposure visits build conviction among CRPs and lead farmers, while FPO linkage turns individual surplus into marketable volume.

Outcome 3: Entitlements that sustain the transition

PVTG households secure entitlements that sustain the transition: FRA and habitat-rights claims facilitated, NMNF enrolment and incentive, PM-JANMAN services and CIG-to-FPO linkage.

This outcome makes the first two last. Secure tenure justifies long-term investment in soil. The NMNF incentive covers households during the early seasons of conversion, when risk is highest. PM-JANMAN services address needs such as water and housing. FPO membership provides a lasting market channel. The targets are at least 600 households enrolled in NMNF, all eligible unclaimed households supported to file FRA claims, and 100 CIGs linked to an FPO. These will be verified through NMNF enrolment records, Tribal Department and FRA committee records, and FPO membership registers.

The supporting outputs are:

- **Output 3.1:** 8 FRA claim-facilitation camps held, with claims facilitated for all eligible listed households, and habitat-rights processes supported with the Tribal Department.
- **Output 3.2:** 12 quarterly block convergence meetings held, and at least 600 households enrolled in NMNF or PM-JANMAN.
- **Output 3.3:** 4 MEL deliverables completed: household listing, baseline, mid-term review and endline.

Output 3.3 underpins the entire chain. Without the listing, the cohort cannot be confirmed. Without the baseline, no target can be validated. Without the endline, no Chhattisgarh evidence on natural farming will exist. The next chapter describes the activities that produce each of these outputs.

Level	Result statement	Indicator	Baseline	Target	Means of verification
Impact	Improved climate resilience and livelihood security of PVTG households in Durgakondal	% of project households reporting stable or improved farm output in a climate-stress season; change in household farm income	To be established by baseline survey	Improvement over baseline among adopters (magnitude validated after baseline)	Baseline and endline household surveys; IMD seasonal data
Outcome 1	PVTG households adopt natural farming practices	No. and % of enrolled households practising at least 3 core natural farming practices on at least part of their land	To be established by baseline survey	At least 600 households (60% of enrolled)	CRP household tracker; endline survey; field verification sample
Outcome 2	Reduced cultivation costs, higher net farm income and crop diversity	Cash input cost per acre; net farm income; number of crops per farm among adopters	To be established by baseline survey	At least 20% lower cash input cost; at least 25% higher net farm income; at least 1 additional crop per farm (targets validated after baseline)	Baseline, mid-term and endline surveys; farmer cost diaries (sample)
Outcome 3	Households secure entitlements that sustain the transition	No. of households enrolled in NMNF; FRA claims filed or resolved; households linked to PM-JANMAN interventions; CIGs linked to an FPO	To be established by baseline survey	At least 600 households enrolled in NMNF; all eligible unclaimed households supported to file FRA claims; 100 CIGs linked to an FPO	NMNF enrolment records; Tribal Department and FRA committee records; FPO membership registers
Output 1.1	20 CRPs trained and active	CRPs completing ToT and refreshers; CRPs certified as Krishi Sakhis	0	20 trained; at least 15 certified	Training attendance; certification records

Output 1.2	Farmers complete farmer field schools	Farmers attending at least 5 of 6 sessions	0	At least 900 farmers	FFS attendance registers; geo-tagged photos
Output 1.3	Demonstration plots, seed banks and BRCs established	Functional demo plots, seed banks and BRCs	0	20 demo plots; 5 seed banks; 5 BRCs	Asset register; field visit reports
Output 2.1	CIGs formed and equipped	CIGs with registers and shared kits in use	0	100	CIG registers; kit distribution records
Output 2.2	Exposure visits and FPO linkage	Participants completing exposure visits; CIGs aggregating through an FPO	0	40 participants; 100 CIGs linked	Visit reports; FPO sales records
Output 3.1	FRA claim facilitation	Camps held; claims facilitated	To be established by baseline survey	8 camps; claims facilitated for all eligible listed households	Camp reports; claim acknowledgement copies
Output 3.2	Block convergence and scheme enrolment	Convergence meetings held; households enrolled in NMNF or PM-JANMAN	0	12 meetings; at least 600 households enrolled	Meeting minutes; scheme records
Output 3.3	MEL completed	Listing, baseline, mid-term review and endline delivered	0	4 deliverables	Survey reports; mid-term review; endline report

Sustainability

The project is designed so that its results depend on institutions and public systems that continue after it ends, rather than on the project itself. Each element of the theory of change has a named successor, and Q12 is a planned handover rather than a closure.

Knowledge stays with the community

The main asset the project creates is knowledge held by 20 CRPs, preferably PVTG women, living in the habitations they serve. Registering them as NMNF Krishi Sakhis gives the role a place in the public system ^{[S14][S15]}. The planned exit is for CRPs to move into NMNF or Bihan roles. Because the practices are learned by doing in field schools and on demonstration plots, they stay with farmers once learned, and neighbours can observe and copy them.

Collectives become a lasting institution

The 100 CIGs will be federated into an FPO. The 10,000 FPO scheme provides up to ₹18 lakh in management support per FPO and ₹2,000 in matching equity per member, paid to the FPO ^[S19]. This funds the institution without project money. FPO revenue from aggregation then gives members a reason to stay and provides a possible source of funding for extension services.

Infrastructure is community-managed

The 5 BRCs and 5 seed banks will be managed by the community. This follows from how they are set up: CIGs prepare inputs jointly at BRC level from the start, so management passes to people who have already been doing the work. Seed banks renew themselves as farmers return seed after harvest. Seeking NMNF set-up support for BRCs ^[S16] also ties these assets to a public programme.

Entitlements do not expire with the project

FRA titles, and any habitat rights secured through alignment with the planned Kamar title covering Kanker ^{[S9][S7]}, are permanent. Links to PM-JANMAN interventions ^{[S17][S18]} and to the block administration, built over 12 quarterly meetings, continue after the project. The NMNF incentive of ₹4,000 per acre per year for 2 years supports households through the early years of conversion ^[S14]. By the time it ends, lower cash input costs are expected to maintain the practice on their own.

Financial logic

Lower input costs are the sustainability mechanism at household level. The Andhra Pradesh evidence showed income gains driven mainly by reduced input spending ^[S10]. A farmer who has stopped buying inputs she no longer needs does not have to be persuaded again each season. The endline will show whether this holds in Durgakondal.

Risk to continuity and mitigation

The main risk is that NMNF is approved only until 2025-26. If it is not extended, the Krishi Sakhi pathway weakens. The design keeps Bihan/SRLM and FPO revenue as parallel routes from the start. The preference for clusters with SRLM groups and FPOs, which comes from NMNF design itself, means that these alternatives are built in rather than added at the last minute. The labour risk is addressed through low-labour sequencing and collective preparation, and the mid-term review in Q6 tests whether these measures are working in time to adjust.

Evidence

The documented baseline, mid-term and endline findings will be the first Chhattisgarh field evidence on natural farming for PVTG households. They will be available to government, SBIF and other practitioners working in similar forested tribal blocks.

Budget

Item	Unit	Unit cost	Qty	Total	Contributes to	Source
A. Personnel & Salaries						
Project Manager (Kanker-based, full time)	person-month	33,000	36	11,88,000	All outcomes; management and convergence	Sector estimate for a remote-block NGO programme manager; an unverified Bihan BPM listing shows about ₹39,875
Natural Farming / Agronomy Specialist	person-month	22,000	36	7,92,000	O1, O2; CRP training and demo plots	Above the BPKP floor of ₹15,000–20,000 honorarium (NITI BPKP guidelines), as a project-level specialist post
Field Coordinators (2; one holds FRA and convergence duties)	person-month	14,000	72	10,08,000	O1–O3; CRP supervision and FRA camps	Above the BPKP LRP floor of ₹10,000; graduate field staff in a remote block
MEL & Data Officer (50% time)	person-month	12,500	36	4,50,000	Output 3.3; logframe tracking	50% of an estimated ₹25,000 per month; no authoritative norm found
Accounts & Admin Assistant (50% time)	person-month	10,000	36	3,60,000	Financial management and compliance	About 50% of an unverified Bihan Accountant/MIS listing of about ₹23,350
CRP / Krishi Sakhi honorarium (20 CRPs, Q2–Q11)	CRP-month	5,000	600	30,00,000	O1, O2; last-mile delivery to about 50 households per CRP	BPKP Guidelines (NITI Aayog Natural Farming portal), ₹5,000 per CRP per month [S20]
Subtotal — Personnel & Salaries				67,98,000		
B. Capital Expenditure						
BRC shed / storage top-up (NMNF ₹1 lakh set-up support sought separately)	per BRC	50,000	5	2,50,000	Output 1.3; community bio-input supply	Low-cost shed estimate; the NMNF norm excludes sheds; PWD schedule of rates or local quote needed
Subtotal — Capital Expenditure				2,50,000		
C. Programme / Activity Costs						
Household listing, baseline and endline surveys	household-round	175	2,000	3,50,000	Output 3.3; target verification	Enumerator cost per household interview; must be at least the state minimum wage

CRP residential ToT (6 days) and refreshers (4 × 2 days)	participant-day	400	280	1,12,000	Output 1.1	MANAGE STRY/ATMA cost norms, ₹400 per trainee per day residential [S21]
CRP-led farmer field school sessions (6 per farmer)	farmer-session	100	6,000	6,00,000	Output 1.2; O1	BPKP Guidelines, ₹100 per farmer per CRP-led village training [S20]
Shared bio-input starter kits (drums, tools) per CIG	per CIG kit	2,500	100	2,50,000	Output 2.1; O1	Market estimate for 1–2 drums plus tools; the NMNF ₹4,000/acre is a ceiling reference only
Model demonstration plots	per plot	6,000	20	1,20,000	Output 1.3	Inputs, signage and field-day costs per plot; no NMNF per-plot norm found
Community seed banks (indigenous varieties)	per seed bank	20,000	5	1,00,000	Output 1.3; climate-resilient seed	Storage bins, initial seed stock and registers; no official norm found
FRA claim-facilitation camps and mapping	per camp	12,500	8	1,00,000	Output 3.1; O3	Venue, documentation, GPS mapping and refreshments; no Tribal Department norm found
Block convergence meetings (quarterly)	per meeting	4,000	12	48,000	Output 3.2	Venue and refreshments for a block-level meeting
Exposure visits for CRPs and lead farmers	per participant	2,500	40	1,00,000	Output 2.2	Travel, stay and food for an in-state visit; the BPKP 7-day training with exposure visit is a cross-check
CIG formation and orientation	per CIG	1,000	100	1,00,000	Output 2.1	Registers, stationery and an orientation meeting per CIG
Staff field travel and mobility	month	9,000	36	3,24,000	All outputs; supervision in remote habitations	Fuel and 2-wheeler reimbursement for field staff; local quotes needed
Mid-term review and learning documentation	lump sum	1,00,000	1	1,00,000	Output 3.3	External reviewer days plus a case-study documentation lump sum
Subtotal — Programme / Activity Costs				23,04,000		
D. Administrative / Overheads						

Field office running, Durgakondal (rent, utilities, internet)	month	12,000	36	4,32,000	Operations base	Below the BPKP ₹20,000/month cluster-office norm (which includes a DEO) because data work sits with the MEL officer; rent quote needed
Statutory audit and compliance	year	35,000	3	1,05,000	Donor compliance	Typical CA fee for a project audit; quotes needed
Subtotal — Administrative / Overheads				5,37,000		
Grand Total				98,89,000		

References

- [S1] Durgkondal Taluk Population, Uttar Bastar Kanker (censusindia2011.com, Census 2011; secondary)
<https://www.censusindia2011.com/chhattisgarh/uttar-bastar-kanker/durgkondal-population.html>
- [S2] List of Villages in Durgkondal Tehsil (VillageInfo.in, Census 2011; secondary)
<https://villageinfo.in/chhattisgarh/uttar-bastar-kanker/durgkondal/>
- [S3] NHSRC – Demographic Status of Scheduled Tribe Population From Census Data
<https://nhsrca.org/sites/default/files/2021-05/Demographic%20Status%20of%20Scheduled%20Tribe%20Population%20-%20NHSRC.pdf>
- [S4] Kanker District Population Census 2011 (census2011.co.in)
<https://www.census2011.co.in/census/district/498-kanker.html>
- [S5] PIB – ECI's efforts bring PVTG communities into electoral process
<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2019294>
- [S6] ThePrint (PTI) – Kamar in Dhamtari gets habitat rights
<https://theprint.in/india/vulnerable-tribal-group-kamar-in-chhattisgarhs-dhamtari-district-gets-habitat-rights/1708448/>
- [S7] UNDP India – Empowering PVTGs: Recognition of Habitat Rights in Chhattisgarh
<https://www.undp.org/india/blog/empowering-particularly-vulnerable-tribal-groups-pvtgs-recognition-habitat-rights-chhattisgarh>
- [S8] MoTA – FRA Monthly Progress Report, September 2021
[https://tribal.nic.in/downloads/FRA/MPR/2021/\(D\)%20MPR%20Sep%202021.pdf](https://tribal.nic.in/downloads/FRA/MPR/2021/(D)%20MPR%20Sep%202021.pdf)
- [S9] Ministry of Tribal Affairs – Forest Rights Act
<https://tribal.nic.in/fra.aspx>
- [S10] GIST Impact – Comparative study: natural farming yields, livelihoods and health
<https://www.gistimpact.com/groundbreaking-comparative-study-reveals-natural-farming-leads-for-yields-livelihoods-and-health/>
- [S11] Natural Farming Through a Wide-Angle Lens: TCA Study of APCNF (2023)
https://futureoffood.org/wp-content/uploads/2025/05/apcnf-tca-study_2023.pdf
- [S12] Global Alliance for the Future of Food – Natural Farming through a Wide Angle Lens
<https://futureoffood.org/publication-library/natural-farming-through-a-wide-angle-lens/>
- [S13] PMIndia – Launch of National Mission on Natural Farming
https://www.pmindia.gov.in/en/news_updates/launch-of-national-mission-on-natural-farming/
- [S14] PIB – National Mission on Natural Farming (NMNF) press note
<https://www.pib.gov.in/PressNoteDetails.aspx?ModuleId=3&NoteId=155019®=3&lang=2>
- [S15] Vikaspedia – National Mission on Natural Farming
<https://en.vikaspedia.in/viewcontent/schemesall/schemes-for-farmers/national-mission-on-natural-farming>
- [S16] Down To Earth – Guidelines for bio-resource centres
<https://www.downtoearth.org.in/agriculture/centre-releases-guidelines-for-setting-up-bio-resource-centres-to-aid-natural-farming-transition>
- [S17] Deccan Herald (PTI) – Cabinet approves PM-JANMAN
<https://www.deccanherald.com/amp/story/india%2Funion-cabinet-approves-pradhan-mantri-jan-jati-advansi-nyaya-maha-abhiyan-2789333>
- [S18] PIB – Rapid Approval of Houses under PM-JANMAN
<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2115300®=48&lang=2>

[S19] PIB – Formation and Promotion of 10,000 FPOs

<https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1696547®=3&lang=2>

[S20] Guidelines of BPKP (NITI Aayog Natural Farming portal)

https://naturalfarming.niti.gov.in/wp-content/uploads/2021/09/Guidelines_of_BPKP.pdf

[S21] MANAGE – STRY & Farmers' Capacity Assessment Cost Norms

<https://www.manage.gov.in/stry&fcac/costnorms-stry.asp>

[S22] DD News – NMNF brings sustainable shift in agri sector

<https://ddnews.gov.in/en/national-mission-on-natural-farming-brings-sustainable-shift-in-agri-sector/>

Rooted Resilience: Natural Farming for Climate Resilience and Livelihood Security of PVTG Households in Durgakondal Block, Uttar Bastar Kanker, Chhattisgarh

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